XIAMEN UNIVERSITY MALAYSIA

ONLINE QUIZ1 EXAMINATION

Course Code:	EEE313			
Course Name:	Semiconductor Physics and Device Processes			
Question Paper Setter:	MohammadMahdi Ariannejad			
Academic Session:	2021/09	Question Paper:	A ☐	B ☐
Total No. of Pages:	1	Time Allocated:	1 hour	
Additional Materials:	Appendix, Calculator			
Apparatus Allowed:	-			

INSTRUCTIONS TO CANDIDATES

1. There are three (3) questions. You are assigned to answer all questions.

2. Each question should be answered in detail with proper Solution, only last answer or only formula won't award any marks.

3. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you would be penalized accordingly.

(Student ID: _____ Full Name: _____)

Question 1: (30 points)

Consider a uniformly doped silicon pn junction with doping concentrations $N_a = 8 \times 10^{17}$ cm^{-3} and $N_d = 12 \times 10^{16}$ cm^{-3}. (*a*) find V_{bi} at $T = 300$ K. (*b*) find the temperature at which V_{bi} increases by 1.6 percent. (Trial and error may have to be used.) ($E_g = 1.12$ eV)

Question 2: (40 Marks)

(*a*) Estimate the total number (# / cm^3) of energy states in silicon between E_c and $E_c + 5kT$ at (*i*) $T = 350$ K and (*ii*) $T = 500$ K. (*b*) Repeat part (*a*) for GaAs.

Question 3: (30 Marks)

Wong is doing revision for his final exam in EEE313. He noticed that in semiconductors, there is some definition based on one-sided p^+n silicon diode. In this condition for silicon material, the doping profile and concentration has been defined in some reference book as of $N_a = 6 \times 10^{14}$ cm^{-3} and $N_d = 12 \times 10^{13}$ cm^{-3} (in room temperature). Now Wong noticed he also needs to know the cross-sectional area which is $A = 8 \times 10^{-8}$ cm^2. ($\tau_{n0} = \tau_{p0} = \tau_0 = 10^{-7}s$, $D_n = 25$ cm^2 and $D_P = 10$ cm^2),

a) Assume that the maximum diffusion capacitance limited to 800 µf

i) Estimate the maximum current in this structure diode
ii) Find the maximum forward-bias voltage
iii) Find the diffusion resistance

b) Now assume that the maximum diffusion capacitance limited to 888 nF

i) Estimate the maximum current in this structure diode
ii) Find the maximum forward-bias voltage
iii) Find the diffusion resistance

- END OF PAPER –

Appendix

Table 1: Important formulas and values for Quiz 1.

Planck constant, h =6.625×10^{-34} J·s= 4.14×10^{-15} eV·s
Free electron mass, m_o = 9.11×10^{-31} kg , $m_n^* = 1.08m_o$ (Si), $m_n^* = 0.067m_o$ (GaAs)
kT=0.0259 eV
Boltzmann constant, k = 8.62×10^{-5} eV/K
1 eV=1.6×10^{-19} J
f(αa) =P'(sin αa)/αa +cosαa=Coska, where $\alpha^2 = \frac{2mE}{\hbar^2}$
$E = \frac{p^2}{2m} = \frac{k^2\hbar^2}{2m}$
$\frac{N(E)}{g(E)} = f_F(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{kT}\right)}$
$g_v(E) = \frac{4\pi(2m_p^*)^{3/2}}{h^3}\sqrt{E_v - E}$
$g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3}\sqrt{E - E_c}$
$J_{diff} = eD_n\frac{dn}{dx} = eD_n\frac{dN_d(x)}{dx}$, $D_n = \mu_n\left(\frac{kT}{e}\right)$
$\rho = \frac{1}{\sigma} = \frac{1}{e(\mu_n n + \mu_p p)}$
$R_n' = R_p' = \frac{\delta n(t)}{\tau_{n0}}$ $\qquad$ $R_n = R_p' = \frac{\delta n(t)}{\tau_{p0}}$
$D_n\frac{\partial^2(\delta n)}{\partial x^2} + \mu_n\left(\mathrm{E}\frac{\partial(\delta n)}{\partial x} + n\frac{\partial \mathrm{E}}{\partial x}\right) + g_n - \frac{n}{\tau_{nt}} = \frac{\partial(\delta n)}{\partial t}$
$C_d = \left(\frac{1}{2V_t}\right)(I_{p0}\tau_{p0} + I_{n0}\tau_{n0})$
$V_{bi} = \frac{kT}{e}\ln\left(\frac{N_aN_d}{n_i^2}\right) = V_t\ln\left(\frac{N_aN_d}{n_i^2}\right)$
$I_D = I_s\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]$ $\qquad$ $I_{n0} = \frac{eAD_n n_{p0}}{L_n}\exp\left(\frac{eV_0}{kT}\right)$
$C' = \left\{\frac{e\epsilon_s N_aN_d}{2(V_{bi} + V_R)(N_a + N_d)}\right\}^{1/2}$

Table 2: Effective Density

Effective density of states function and density of states effective mass values

	N_c (cm^{-3})	N_v (cm^{-3})	m_n^*/m_0	m_p^*/m_0
Silicon	2.8×10^{19}	1.04×10^{19}	1.08	0.56
Gallium arsenide	4.7×10^{17}	7.0×10^{18}	0.067	0.48
Germanium	1.04×10^{19}	6.0×10^{18}	0.55	0.37

Table 3: values for n_i

Commonly accepted values of n_i at $T = 300$ K	
Silicon	$n_i = 1.5 \times 10^{10}\ \text{cm}^{-3}$
Gallium arsenide	$n_i = 1.8 \times 10^{6}\ \text{cm}^{-3}$
Germanium	$n_i = 2.4 \times 10^{13}\ \text{cm}^{-3}$